UNCONDITIONAL HYPERBOLICITY OF THE CRITICAL LAGUERRE FAMILY: AN ELEMENTARY AND FORMALLY VERIFIED PROOF

MICHAEL HAUFSCHILD

ABSTRACT. For $d \geq 1$ let $C_d(w) = {}_1F_1\big(-d;\, \tfrac{1}{2};\, w^2/4d\big)$ — up to normalization the Laguerre polynomial $L_d^{(-1/2)}$ at its critical scaling — and let

$$\Psi_d(w) \;=\; \sum_{k=0}^{d}(-1)^k \,\frac{d^{\underline{k}}\, M_k}{d^{\,k}\,(2k)!}\, w^{2k}, \qquad M_k = \frac{(2k)!}{4^k\, k!}\, \mathrm{e}^{\gamma k - S_1(k)},$$

where $d^{\underline{k}}$ is the falling factorial, γ is Euler's constant and $S_1(k) = \sum_{j=1}^{k} H_{j-1}$ is the partial sum of harmonic numbers. We prove that for every $d \geq 1$ all $2d$ zeros of Ψ_d are real and simple. The numbers M_k are shown to be the moment sequence of an explicit compound-Poisson random variable $U \in (0, 1]$ with atom $\mathbb{P}(U = 1) = \sqrt{2/\mathrm{e}}$, so that $\Psi_d(w) = \mathbb{E}\, C_d(\sqrt{U}\, w)$: hyperbolicity of the whole family is an exact averaging, not an asymptotic, phenomenon. The proof is elementary — real-variable throughout, using only the Kummer differential equation, an energy monotonicity along the real axis, and a sign-alternation count at the critical points of C_d — and it has been formally verified: the theorem is proved in Lean 4 against the mathlib library with zero `sorry`s, depending only on the three standard axioms. The sequence (M_k) is provably *not* a Pólya–Schur multiplier sequence — no nondegenerate moment sequence is, since moment sequences are log-convex while multiplier sequences must be log-concave — so the hyperbolicity proved here is a property of the pair (sequence, family) and is not subsumed by any universal coefficientwise preserver theorem. To our knowledge Ψ_d is, in this generality, a new explicit hyperbolic deformation of the Laguerre–Hermite family.

1. INTRODUCTION

A real polynomial is *hyperbolic* if all of its complex zeros are real. The hyperbolicity of the classical orthogonal families — Hermite, Laguerre, Jacobi — is classical [1, Thm. 3.3.1], and the linear operations preserving hyperbolicity were characterized by Pólya and Schur [2]. Deciding hyperbolicity of a *specific* explicitly given family that is not covered by such preserver theorems remains, in general, a genuinely hard problem: one must either locate all the zeros or manufacture enough sign changes.

Date: June 12, 2026.

This paper proves hyperbolicity, with simple zeros, for the following family. Throughout, $d^{\underline{k}} = d(d-1)\cdots(d-k+1)$ denotes the falling factorial, $H_n = \sum_{j=1}^{n} 1/j$ the harmonic numbers ($H_0 = 0$), γ Euler's constant, and

$$S_1(k) \;=\; \sum_{j=1}^{k} H_{j-1} \;=\; kH_{k-1} - (k-1) \qquad (k \geq 1), \qquad S_1(0) = 0. \tag{1}$$

Definition 1.1. For integers $k \geq 0$ and $d \geq 1$ set

$$M_k \;=\; \frac{(2k)!}{4^k\, k!}\, \mathrm{e}^{\gamma k - S_1(k)}, \tag{2}$$

$$\begin{aligned} C_d(w) \;&=\; \sum_{k=0}^{d} (-1)^k \frac{d^{\underline{k}}}{d^{\,k}(2k)!}\, w^{2k} \;=\; {}_1F_1\!\left(-d;\, \tfrac{1}{2};\, \frac{w^2}{4d}\right), \\ \Psi_d(w) \;&=\; \sum_{k=0}^{d} (-1)^k \frac{d^{\underline{k}}\, M_k}{d^{\,k}(2k)!}\, w^{2k}. \end{aligned} \tag{3}$$

By Kummer's transformation of the confluent hypergeometric polynomial, C_d is the Laguerre polynomial $L_d^{(-1/2)}$ at the argument scaling $w^2/4d$ that places its zeros at unit mean density — we call $\{C_d\}_{d\geq 1}$ the *critical Laguerre family*, and Ψ_d its *moment deformation* by the sequence (2). The deformation is of genuinely infinite order: the sequence (M_k) is strictly decreasing (the ratio test below reduces this to $\log(1+x) < x$), from $M_0 = 1$ through $M_1 = \mathrm{e}^{\gamma}/2 = 0.89053\ldots$ toward its limit $p := \sqrt{2/\mathrm{e}} = 0.85776\ldots$, so every coefficient of C_d is damped by a different factor. Moreover, (M_k) is provably *not* a multiplier sequence in the sense of Pólya–Schur: already the image of $(w^2-1)^2$ under the coefficientwise action has four nonreal zeros (Proposition 6.3), so no universal coefficientwise preserver theorem for all even hyperbolic polynomials can yield Theorem 1.2 — the hyperbolicity below is a property of the pair (sequence, family), not of the sequence alone.

Theorem 1.2 (Main theorem)**.** *For every integer $d \geq 1$, all $2d$ zeros of Ψ_d are real and simple.*

The proof has two ingredients, both of independent interest.

First ingredient: the moments are a compound-Poisson law. The sequence (2) is exactly the moment sequence of an explicit random variable: let N be a Poisson point process on $(0,\infty)$ with the finite intensity measure

$$\lambda(\,\mathrm{d}t) \;=\; \eta(t)\,\frac{\mathrm{e}^{-t}}{1-\mathrm{e}^{-t}}\,\mathrm{d}t, \qquad \eta(t) \;=\; \frac{\mathrm{e}^{t/2}}{t} - \frac{1}{1-\mathrm{e}^{-t}} \;>\; 0,$$

and set $U = \mathrm{e}^{-\sum_j T_j}$, the product over the points of N. Then (Theorem 2.4)

$$\mathbb{E}\,[U^k] \;=\; M_k \quad (k \geq 0), \qquad \mathbb{P}(U = 1) \;=\; p \;=\; \sqrt{2/\mathrm{e}}\,.$$

Consequently, writing ν for the law of $v = \sqrt{U}$ and $\mu = \nu - p\,\delta_1$ for its non-atomic part,

$$\Psi_d(w) \;=\; \mathbb{E}\, C_d\big(\sqrt{U}\, w\big) \;=\; p\, C_d(w) \;+\; \int_{(0,1]} C_d(vw)\; \mathrm{d}\mu(v), \tag{4}$$

with $\mu \geq 0$ and $\mu\big((0,1]\big) = 1 - p$. The identity (4) is exact, coefficient by coefficient, at every finite d; no limit is involved. Nor is the intensity λ an ansatz: Section 2 opens by deriving it from the moment ratios of (2), through the classical integral representations of the digamma function.

Second ingredient: a one-line energy bound at the critical points. Writing $E = C_d^2 + C_d'^2$, the Kummer differential equation $C_d'' - (w/2d)\, C_d' + C_d = 0$ gives $E'(x) = (x/d)\, C_d'(x)^2 \geq 0$ on $x \geq 0$; since $E(0) = 1$, every critical point c of C_d satisfies $|C_d(c)| \geq 1$ and, crucially, the *whole segment* $[0, c]$ is dominated: $|C_d(vc)| \leq |C_d(c)|$ for $0 \leq v \leq 1$. Feeding this into (4) and using only $p > \frac{1}{2}$ shows that Ψ_d *copies the sign pattern* of C_d at the critical points of C_d. The classical interlacing data of the Laguerre zeros then forces d sign changes of Ψ_d on $(0, \infty)$, which (with evenness and the degree count) pins all $2d$ zeros to the real axis, each simple.

No complex analysis enters: no Rouché contour, no Hermite–Biehler theory, no saddle points. The argument runs entirely on the real line, which is also what made a complete formal verification practical.

Formal verification. Theorem 1.2 has been proved in Lean 4 [6] against the mathlib library [7]: the statement

```
theorem theorem_M (d : ℕ) (hd : 1 ≤ d) :
∀ z ∈ ((Psi d).map (algebraMap ℝ ℂ)).roots, z.im = 0
```

compiles with zero `sorry`s and depends only on the three standard axioms of the Lean/mathlib universe (`propext`, `Classical.choice`, `Quot.sound`). Section 5 describes the formalization; the complete source, with build instructions and the verification scripts, is available at

`https://github.com/michael-haufschild-gib/theorem-m`

Motivation. The identity (4) exhibits Ψ_d as an exact average of rescaled copies of C_d, and Theorem 1.2 says that this particular averaging preserves hyperbolicity at every degree. Read as a statement about the sequence (M_k), it is a concrete instance of a natural question in the tradition of Pólya–Schur preserver theory: *which moment sequences of random variables in* $[0, 1]$ *act coefficientwise on the critical Laguerre family without destroying hyperbolicity?* The question is intrinsically tied to the family: no nondegenerate moment sequence preserves hyperbolicity of *all* even hyperbolic polynomials (Remark 6.4 — moment sequences are log-convex, multiplier sequences must be log-concave), so family-specific structure is the only possible source of such preservation. The compound-Poisson mechanism of Section 2 — an atom of mass $p > \frac{1}{2}$ at the identity scale plus a positive residual — combined with the energy bound of Section 3 yields a sufficient criterion of striking simplicity, and the sequence (2), with its digamma–Binet moment structure and atom exactly $\sqrt{2/\mathrm{e}}$, is a distinguished instance in which the criterion is endpoint-sharp (Remark 6.1). Beyond that, the theorem stands on its own: an explicit, nontrivially deformed Laguerre family that remains hyperbolic at every degree, with an elementary, machine-checked proof and an exact probabilistic mechanism behind it.

Organization. Section 2 constructs the compound-Poisson law and proves $\mathbb{E}[U^k] = M_k$. Section 3 collects the classical facts about C_d and proves the energy lemma. Section 4 proves Theorem 1.2. Section 5 documents the formal verification. Section 6 closes with remarks on sharpness and a proof that (M_k) is *not* a Pólya–Schur multiplier sequence, so that no universal coefficientwise preserver theorem subsumes Theorem 1.2.

2. The compound-Poisson law behind the moments

Reverse-engineering the intensity. Before verifying anything, we explain where λ comes from: it is forced by the moments, and the harmonic numbers in (2) dictate its shape through the digamma function. The ratios of consecutive moments are elementary — by the telescoping $S_1(k) - S_1(k-1) = H_{k-1}$ of (1),

$$\frac{M_k}{M_{k-1}} = \frac{2k(2k-1)}{4k}\,\mathrm{e}^{\gamma - H_{k-1}} = \Big(k - \tfrac{1}{2}\Big)\,\mathrm{e}^{\gamma - H_{k-1}}.$$

Now look for a random variable $U = \mathrm{e}^{-Y} \in (0,1]$ with these moments, taking $Y \geq 0$ infinitely divisible — the simplest class with fully explicit Laplace transforms: $\log \mathbb{E}[\mathrm{e}^{-kY}] = \int_0^\infty (\mathrm{e}^{-kt} - 1)\,\lambda(\mathrm{d}t)$ for an intensity measure λ to be found. Consecutive log-moment differences are then $\int (\mathrm{e}^{-kt} - \mathrm{e}^{-(k-1)t})\,\lambda(\mathrm{d}t)$, and the ansatz $\lambda(\mathrm{d}t) = \eta(t)\,\frac{\mathrm{e}^{-t}}{1-\mathrm{e}^{-t}}\,\mathrm{d}t$ makes the difference collapse to a single Laplace transform, since $(\mathrm{e}^{-kt} - \mathrm{e}^{-(k-1)t})\,\frac{\mathrm{e}^{-t}}{1-\mathrm{e}^{-t}} = -\mathrm{e}^{-kt}$. Matching the moment ratios above therefore amounts to solving

$$\int_0^\infty \mathrm{e}^{-kt}\,\eta(t)\,\mathrm{d}t \;=\; \psi(k) - \log\Big(k - \tfrac{1}{2}\Big), \qquad k = 1, 2, \ldots,$$

where the harmonic numbers have become the digamma function through $\psi(k) = H_{k-1} - \gamma$. This transform pair is classical: Gauss's integral represents ψ, Frullani's integral represents the logarithm (the half-integer shift $k - \frac{1}{2}$ producing the factor $\mathrm{e}^{t/2}$), and subtracting the two integrands term by term reads off

$$\eta(t) \;=\; \frac{\mathrm{e}^{t/2}}{t} - \frac{1}{1-\mathrm{e}^{-t}}.$$

Everything to this point is bookkeeping. The one substantive fact — the fact that makes the random variable exist and the whole method work — is that this particular difference of classical integrands is *positive* and integrable against $\frac{\mathrm{e}^{-t}}{1-\mathrm{e}^{-t}}\,\mathrm{d}t$. That is Lemma 2.1, and the rest of the section re-runs the derivation forwards, rigorously.

Lemma 2.1. *For $t>0$ let $\eta(t)=\dfrac{\mathrm{e}^{t/2}}{t}-\dfrac{1}{1-\mathrm{e}^{-t}}$. Then*

$$\eta(t) \;=\; \frac{2\sinh(t/2)-t}{t\,(1-\mathrm{e}^{-t})} \;>\; 0, \qquad \eta(t)=\frac{t}{24}+O(t^2)\quad(t\downarrow 0),$$
$$\eta(t)\,\frac{\mathrm{e}^{-t}}{1-\mathrm{e}^{-t}} = O\big(t^{-1}\mathrm{e}^{-t/2}\big)\quad(t\to\infty).$$

In particular $\lambda(\,\mathrm{d}t) := \eta(t)\,\frac{\mathrm{e}^{-t}}{1-\mathrm{e}^{-t}}\,\mathrm{d}t$ is a finite positive measure on $(0,\infty)$.

Proof. Multiplying η by $t(1-\mathrm{e}^{-t})>0$ gives $\mathrm{e}^{t/2}(1-\mathrm{e}^{-t})-t=\mathrm{e}^{t/2}-\mathrm{e}^{-t/2}-t=2\sinh(t/2)-t>0$ for $t>0$, since $\sinh u>u$ for $u>0$. The expansion at 0 is immediate from $2\sinh(t/2)-t=t^3/24+O(t^5)$ and $t(1-\mathrm{e}^{-t})=t^2+O(t^3)$; the decay at infinity from $2\sinh(t/2)\le \mathrm{e}^{t/2}$. Both endpoints are integrable against $\mathrm{e}^{-t}/(1-\mathrm{e}^{-t})\,\mathrm{d}t\sim\,\mathrm{d}t/t$ at 0 and $\sim\mathrm{e}^{-t}\,\mathrm{d}t$ at ∞, whence λ is finite. □

Definition 2.2. Let $N=\{T_j\}$ be a Poisson point process on $(0,\infty)$ with intensity λ (a.s. finitely many points, since λ is finite), and set

$$Y \;=\; \sum_j T_j \;\in\; [0,\infty), \qquad U \;=\; \mathrm{e}^{-Y} \;\in\; (0,1].$$

The variable U is a compound-Poisson exponential functional; its atom at 1 is the no-point event, $\mathbb{P}(U=1)=\mathrm{e}^{-\lambda((0,\infty))}>0$. By Campbell's theorem (the Laplace functional of a Poisson process, [5, Sec. 3.2]),

$$\mathbb{E}\,[U^k] \;=\; \mathbb{E}\,\mathrm{e}^{-kY} \;=\; \exp\Big(\int_0^\infty \big(\mathrm{e}^{-kt}-1\big)\lambda(\,\mathrm{d}t)\Big), \qquad k\ge 0. \tag{5}$$

The transform of η is a Binet-type integral. We only need it at the positive integers, where it is elementary:

Lemma 2.3 (integer Binet identity)**.** *For every integer $k\ge 1$,*

$$\int_0^\infty \mathrm{e}^{-kt}\,\eta(t)\,\mathrm{d}t \;=\; H_{k-1}-\gamma-\log\Big(k-\tfrac12\Big).$$

Proof. Split $\eta(t)\mathrm{e}^{-kt}=\big(\mathrm{e}^{-(k-1/2)t}-\mathrm{e}^{-t}\big)/t+\big(\mathrm{e}^{-t}/t-\mathrm{e}^{-kt}/(1-\mathrm{e}^{-t})\big)$. Each part is absolutely integrable: the first is bounded near 0 (numerator $=O(t)$) and decays exponentially; the second is Gauss's integrand for the digamma function. Frullani's integral [4, Sec. 12.16] gives

$$\int_0^\infty \frac{\mathrm{e}^{-(k-1/2)t}-\mathrm{e}^{-t}}{t}\,\mathrm{d}t=-\log\Big(k-\tfrac12\Big),$$

and Gauss's formula [4, Sec. 12.3]

$$\int_0^\infty \Big(\frac{\mathrm{e}^{-t}}{t}-\frac{\mathrm{e}^{-kt}}{1-\mathrm{e}^{-t}}\Big)\,\mathrm{d}t=\psi(k)=H_{k-1}-\gamma$$

for the digamma function $\psi=\Gamma'/\Gamma$ at the positive integer k. Adding the two evaluations gives the claim. □

Theorem 2.4 (moment identification)**.** $\mathbb{E}\,[U^k] = M_k$ *for every integer* $k \geq 0$, *and* $\mathbb{P}(U = 1) = p = \sqrt{2/\mathrm{e}}$. *Consequently, with* ν *the law of* $v = \sqrt{U}$ *and* $\mu := \nu - p\,\delta_1$, μ *is a positive measure concentrated on* $(0,1]$ *with total mass* $1-p$, *and for every* $d \geq 1$ *the exact decomposition* (4) *holds.*

Proof. Ratios. For $k \geq 1$, by (5),

$$\log \frac{\mathbb{E}[U^k]}{\mathbb{E}[U^{k-1}]} = \int_0^\infty \left(\mathrm{e}^{-kt} - \mathrm{e}^{-(k-1)t}\right) \eta(t)\, \frac{\mathrm{e}^{-t}}{1-\mathrm{e}^{-t}}\ \mathrm{d}t = -\int_0^\infty \mathrm{e}^{-kt}\, \eta(t)\ \mathrm{d}t,$$

because $\left(\mathrm{e}^{-kt} - \mathrm{e}^{-(k-1)t}\right)\mathrm{e}^{-t}/(1-\mathrm{e}^{-t}) = -\mathrm{e}^{-kt}$. By Lemma 2.3,

$$\frac{\mathbb{E}[U^k]}{\mathbb{E}[U^{k-1}]} = \left(k - \tfrac{1}{2}\right) \mathrm{e}^{\gamma - H_{k-1}}. \tag{6}$$

On the other hand, from (2) and the telescoping $S_1(k) - S_1(k-1) = H_{k-1}$ of (1),

$$\frac{M_k}{M_{k-1}} = \frac{2k(2k-1)}{4k}\, \mathrm{e}^{\gamma - H_{k-1}} = \left(k - \tfrac{1}{2}\right) \mathrm{e}^{\gamma - H_{k-1}}.$$

Since $\mathbb{E}[U^0] = 1 = M_0$, induction gives $\mathbb{E}[U^k] = M_k$ for all k.

Atom. Since $U \in (0,1]$, $U^k \downarrow \mathbf{1}_{\{U=1\}}$ pointwise, so by dominated convergence $\mathbb{P}(U=1) = \lim_k \mathbb{E}[U^k] = \lim_k M_k$. Stirling's formula gives

$$\frac{(2k)!}{4^k\, k!} = \sqrt{2}\, k^k \mathrm{e}^{-k} \left(1 + O(1/k)\right),$$

while $H_{k-1} = \log k + \gamma - \frac{1}{2k} + O(k^{-2})$ and (1) give $\gamma k - S_1(k) = -k\log k + k - \frac{1}{2} + O(1/k)$; multiplying, $M_k = \sqrt{2}\,\mathrm{e}^{-1/2} + O(1/k) \to \sqrt{2/\mathrm{e}}$.

Decomposition. $v = \sqrt{U} \in (0,1]$ has an atom of mass exactly p at 1, so $\mu = \nu - p\delta_1 \geq 0$ is concentrated on $(0,1]$ with mass $1-p$ (in fact $\mu(\{1\}) = 0$, since the atom of ν at 1 has mass exactly p; nothing below uses this, and we do not record it formally). Finally, for any $d \geq 1$, expanding the finite sum (3) and using $\mathbb{E}[(\sqrt{U})^{2k}] = \mathbb{E}[U^k] = M_k$,

$$\mathbb{E}\, C_d(\sqrt{U}\, w) = \sum_{k=0}^{d} (-1)^k \frac{d^{\underline{k}}}{d^{\,k}(2k)!}\, \mathbb{E}[U^k]\ w^{2k} = \Psi_d(w),$$

which is (4) after splitting $\nu = p\,\delta_1 + \mu$. $\square$

Remark 2.5. Only three consequences of Theorem 2.4 are used below: $\mu \geq 0$, the mass $\mu((0,1]) = 1-p$, and the support $\operatorname{supp}\mu \subseteq [0,1]$. The specific shape of λ matters only through the closed form (2) of the moments. Note also $p^2 = 2/\mathrm{e} > 1/4$, i.e.

$$p \ > \ \tfrac{1}{2}, \tag{7}$$

the single inequality on which the entire sign-transfer argument runs (it amounts to $8 > \mathrm{e}$).

3. The critical Laguerre polynomial and its energy

Lemma 3.1 (Hermite bridge and zeros)**.** *Let* He_n *denote the probabilists' Hermite polynomials,* $\mathrm{He}_n(x) = (-1)^n \mathrm{e}^{x^2/2} \frac{\mathrm{d}^n}{\mathrm{d}x^n} \mathrm{e}^{-x^2/2}$*. Then*

$$C_d(w) \;=\; \frac{\mathrm{He}_{2d}\big(w/\sqrt{2d}\big)}{\mathrm{He}_{2d}(0)}, \qquad \mathrm{He}_{2d}(0) = (-1)^d\,(2d-1)!!. \tag{8}$$

Consequently C_d *is an even polynomial of degree* $2d$ *with* $C_d(0) = 1$*, and all its zeros are real and simple:*

$$-\rho_d < \cdots < -\rho_1 < 0 < \rho_1 < \cdots < \rho_d.$$

Proof. The quadratic transformation between Hermite and Laguerre polynomials [1, (5.6.1)], in confluent-hypergeometric form [3, Ch. 18], reads $H_{2d}(x) = (-1)^d \frac{(2d)!}{d!}\, {}_1F_1\big(-d;\, \frac{1}{2};\, x^2\big)$ for the physicists' polynomials H_n; converting by $\mathrm{He}_n(x) = 2^{-n/2} H_n(x/\sqrt{2})$ and substituting $x = w/\sqrt{4d}$ yields (8), the constant being $\mathrm{He}_{2d}(0) = (-1)^d (2d)!/(2^d d!) = (-1)^d (2d-1)!!$. The zeros of Hermite polynomials are real and simple [1, Thm. 3.3.1], and 0 is not among them for even index since $\mathrm{He}_{2d}(0) \neq 0$; evenness of He_{2d} arranges them symmetrically.

Remark on the formal proof. In the Lean formalization the identity (8) is not imported from the literature: both sides are even polynomials with constant term 1 whose even coefficient sequences satisfy the same first-order recurrence

$$(2k+1)(2k+2)\, a_{k+1} \;=\; \big(2k - 2d\big)\, a_k \,/\, (2d)$$

forced by their respective differential equations, and the real-rootedness of He_n, with simplicity and the interlacing used below, is proved from scratch by induction on n through the recurrence $\mathrm{He}_{n+1} = x\mathrm{He}_n - \mathrm{He}_n'$ (Section 5). □

Lemma 3.2 (Kummer ODE)**.** $C_d''(w) - \dfrac{w}{2d}\, C_d'(w) + C_d(w) = 0.$

Proof. With $u(z) = {}_1F_1(-d;\, 1/2;\, z)$ and $z = w^2/4d$, Kummer's equation $z\, u'' + (\frac{1}{2} - z)\, u' + d\, u = 0$ [3, Ch. 13] transforms under $C_d(w) = u(w^2/4d)$ by the chain rule: $C_d' = u' \cdot w/2d$ and $C_d'' = u''\, w^2/4d^2 + u'/2d$, so, collecting terms (all derivatives of u evaluated at $z = w^2/4d$),

$$C_d'' - \frac{w}{2d} C_d' + C_d = \frac{u''\, w^2}{4d^2} + \frac{u'}{2d} - \frac{u'\, w^2}{4d^2} + u = \frac{1}{d}\Big(z\, u'' + \big(\tfrac{1}{2} - z\big) u' + d\, u\Big) \;=\; 0. \quad \square$$

Lemma 3.3 (critical data)**.** C_d' *has exactly* $2d-1$ *zeros, all real and simple; on* $[0, \infty)$ *they are* $0 = c_0 < c_1 < \cdots < c_{d-1}$*, interlacing with the zeros as*

$$0 = c_0 < \rho_1 < c_1 < \rho_2 < \cdots < c_{d-1} < \rho_d,$$

and the critical values alternate in sign:

$$\operatorname{sgn} C_d(c_m) \;=\; (-1)^m, \qquad m = 0, \ldots, d-1. \tag{9}$$

For $d = 1$ *the list of positive critical points is empty and* (9) *reduces to* $C_1(0) = 1 > 0$*.*

Proof. C_d' has degree $2d-1$; Rolle's theorem places at least one of its zeros strictly between consecutive zeros of C_d, which accounts for $2d-1$ of them and therefore for all, each simple, one per gap. The gap $(-\rho_1, \rho_1)$ contains the critical point 0 (evenness). On (ρ_m, ρ_{m+1}) the polynomial C_d has no zero, hence constant sign; the sign on $(-\rho_1, \rho_1)$ is $+$ because $C_d(0) = 1$, and it flips across each simple zero ρ_m, so the sign on (ρ_m, ρ_{m+1}) is $(-1)^m$. Since $c_m \in (\rho_m, \rho_{m+1})$ for $1 \le m \le d-1$ and $c_0 = 0$, the display (9) follows. $\square$

Lemma 3.4 (energy monotonicity and segment domination)**.** *Let* $E(x) = C_d(x)^2 + C_d'(x)^2$. *Then* $E'(x) = \dfrac{x}{d} C_d'(x)^2 \ge 0$ *for* $x \ge 0$, *and* $E(0) = 1$. *Consequently, for every critical point* $c \ge 0$ *of* C_d *and every* $v \in [0,1]$,

$$|C_d(vc)| \;\le\; |C_d(c)|, \qquad \text{and} \qquad |C_d(c)| \;\ge\; 1. \tag{10}$$

Proof. Differentiate and substitute the ODE of Lemma 3.2:

$$E' = 2C_d' \left(C_d + C_d''\right) = 2C_d' \cdot \frac{x}{2d} C_d' = \frac{x}{d} C_d'^2 \;\ge\; 0 \qquad (x \ge 0).$$

$E(0) = C_d(0)^2 + C_d'(0)^2 = 1 + 0$. Monotonicity gives, for $0 \le v \le 1$,

$$C_d(vc)^2 \;\le\; E(vc) \;\le\; E(c) \;=\; C_d(c)^2,$$

the last equality because $C_d'(c) = 0$; and $C_d(c)^2 = E(c) \ge E(0) = 1$. $\square$

Remark 3.5. Lemma 3.4 is the real-axis case of an energy monotonicity that in fact holds along every horizontal line in $\mathbb{C}$ ($\partial_x(|C_d|^2 + |C_d'|^2) = (x/d)|C_d'|^2$, the imaginary part of w contributing no real part to the cross terms). Only the real-axis case is needed here, and only it was formalized. We also note that (10) with constant 1 is what makes the argument work with no room to spare; see Remark 6.1.

4. Proof of the main theorem

Proof of Theorem 1.2. Fix $d \ge 1$ and abbreviate $p = \sqrt{2/\mathrm{e}}$. Recall from (4) that $\Psi_d(x) = p\, C_d(x) + \int_{(0,1]} C_d(vx)\, \mathrm{d}\mu(v)$ with $\mu \ge 0$ of total mass $1-p$.

Step 1: Ψ_d copies the critical signs of C_d. Let $c \in \{c_0, \dots, c_{d-1}\}$ be a nonnegative critical point of C_d. By the segment domination (10),

$$\Big|\Psi_d(c) - p\, C_d(c)\Big| \;\le\; \int_{(0,1]} \big|C_d(vc)\big|\, \mathrm{d}\mu(v) \;\le\; (1-p)\, \big|C_d(c)\big| \;<\; p\, \big|C_d(c)\big|,$$

the last (strict) inequality by (7) and $|C_d(c)| \ge 1 > 0$. Hence $\Psi_d(c) \ne 0$ and

$$\operatorname{sgn} \Psi_d(c_m) \;=\; \operatorname{sgn} C_d(c_m) \;=\; (-1)^m, \qquad m = 0, \dots, d-1. \tag{11}$$

Step 2: tail sign. The leading coefficient of Ψ_d is $(-1)^d\, d!\, M_d / \big(d^{\,d} (2d)!\big)$, of sign $(-1)^d$ since $M_d > 0$; the degree is exactly $2d$. Hence $\operatorname{sgn} \Psi_d(x) = (-1)^d$ for all sufficiently large $x > 0$.

Step 3: count. By (11) and Step 2 the sequence of values

$$\Psi_d(c_0),\ \Psi_d(c_1),\ \dots,\ \Psi_d(c_{d-1}),\ \Psi_d(x_\infty) \qquad (x_\infty \text{ large})$$

has signs $+, -, +, \ldots, (-1)^{d-1}, (-1)^d$: it alternates through d strict sign changes. By the intermediate value theorem Ψ_d has at least d distinct zeros in $(c_0, \infty) = (0, \infty)$. Since Ψ_d is even, it has at least d further distinct zeros in $(-\infty, 0)$, and $\Psi_d(0) = M_0 = 1 \neq 0$. That is at least $2d$ distinct complex zeros of a polynomial of degree $2d$: there are exactly $2d$, all real, all simple. □

Remark 4.1 ($d = 1$)*.* The degenerate case is instructive: there are no positive critical points, the sign sequence is $\Psi_1(0) = 1 > 0$ followed by the tail sign $-$, and indeed $\Psi_1(w) = 1 - \frac{e^\gamma}{4} w^2$ has the two real simple zeros $\pm 2\,e^{-\gamma/2}$. The proof above covers this case verbatim — no step assumed $d \geq 2$.

5. The formal proof

Theorem 1.2 (in the realness form stated in Section 1) has been formalized in Lean 4 (v4.30.0) against mathlib (pinned at commit `c5ea0035` / tag v4.30.0).[1] The development comprises thirteen files, about 5,300 lines:

`Defs`	the polynomials Ψ_d, C_d, the sequences M_k, S_1
`Structure`	coefficient closed forms, degree, the Kummer ODE in polynomial form
`Frullani`	Frullani's integral for exponentials
`Binet`	the integer Binet identity (Lemma 2.3)
`CPMeasure`	the Lévy measure λ as a `withDensity` measure; finiteness
`Moments`, `MomentsLimit`	moment ratios; the Stirling limit $M_k \to p$
`MuBridge`	the finite-quadrature interface: residual moments $M_k - p$ from a measure
`Capstone`	discharge of the quadrature interface by the compound-Poisson pushforward
`Energy`	Lemma 3.4 (real energy monotonicity, segment domination)
`Hermite`	real-rootedness, simplicity and interlacing of He_n, by induction
`CriticalData`	Lemma 3.3 for C_d via the bridge (8); statement and proof of `theorem_M` and `theorem_M_aeval`
`SignCount`	the sign-alternation count, Step 3 above

Two design points deserve mention.

Measures, not samples. The formal development verifies every measure-level assertion of Section 2: positivity and finiteness of λ (a `withDensity` measure), the integer Binet identity as a genuine Lebesgue-integral statement, and the existence of the residual measure μ of (4) — obtained as a pushforward — with $\mu \geq 0$, concentration on $(0, 1]$, mass $1 - p$ and even moments $M_k - p$. What it does not do is represent U as the exponential functional $e^{-\sum_j T_j}$ of a Poisson point process. This is a deliberate boundary, not a gap: the proof of Theorem 1.2 consumes the law of U only through the listed measure data, so the point-process representation — the right way to *understand* the law, and the way it was found (Section 2) — contributes no

[1]Source, build instructions and verification scripts: `https://github.com/michael-haufschild-gib/theorem-m`.

additional logical content to the theorem. Formalizing it would enlarge the verified statement about the measure, not the verified theorem about Ψ_d.

Hermite theory from scratch. Rather than formalize the classical literature on Laguerre zeros, the development proves real-rootedness with simplicity and the full interlacing data for the probabilists' Hermite polynomials by induction on n via $\mathrm{He}_{n+1} = x\,\mathrm{He}_n - \mathrm{He}_n'$: between consecutive simple roots of He_n the derivative term forces a sign change of He_{n+1}, and two tail roots are produced by the parity of the leading terms; a counting argument then pins exactly $n+1$ simple roots. The identity (8) transports this package to C_d by a first-order coefficient recurrence, avoiding any appeal to unformalized special function theory.

The final check reports, for the headline theorem and its evaluation-form companion alike,

```
'TheoremM.theorem_M' depends on axioms:
    [propext, Classical.choice, Quot.sound]
'TheoremM.theorem_M_aeval' depends on axioms:
    [propext, Classical.choice, Quot.sound]
```

with zero `sorry`s in the tree (8489 compiled jobs). These three axioms are the standard foundation of all of mathlib; in particular no numerical computation, floating point or interval arithmetic enters the formal proof anywhere — all constants, including $p = \sqrt{2/\mathrm{e}}$ and the inequality (7), are handled exactly.

The formalization was produced jointly by two AI research agents (Claude "Fable" and GPT) working under the author's direction in a mutual-audit protocol (disjoint file ownership, cross line-audits of each other's proofs), with the Lean kernel as the final referee. In addition, the development has been re-verified independently of Lean's own kernel: the proof terms were exported with the upstream `lean4export` tool (v4.30.0) and re-checked by Nanoda (`nanoda_lib`, commit `f58f2f6d`), an independent implementation of the Lean 4 kernel in Rust,[2] which reports *"Checked 48241 declarations with no errors"* for an export containing `theorem_M` and `theorem_M_aeval`, with the permitted axioms restricted to exactly the three listed above.

6. Concluding remarks

Remark 6.1 (sharpness)*.* The budget of Step 1 is sharp at the endpoint $v \to 1^-$: numerically, for every d tested ($1 \le d \le 300$), the worst ratio $|\Psi_d(c_m) - pC_d(c_m)| \,/\, p|C_d(c_m)|$ over the critical points equals $(1-p)/p = 0.16582\ldots$ to all computed digits — the measure μ concentrates its action at v near 1, where the domination (10) costs nothing. The proof therefore has a uniform factor-6 margin, but no asymptotic slack is hiding anywhere: the mechanism is the strict inequality $p > \frac{1}{2}$, used once.

[2] `https://github.com/leanprover/lean4export` and `https://github.com/ammkrn/nanoda_lib`.

Remark 6.2 (simplicity vs. the formal statement). The paper proof gives realness *and simplicity* (Step 3 produces $2d$ *distinct* zeros). The formalized statement asserts realness of every complex root; distinctness is present in the formal development's counting layer but was not exported into the headline statement.

The moment sequence (2) acts coefficientwise on C_d and preserves hyperbolicity for every d. It is natural to ask whether $(M_k)_{k\geq 0}$ is a Pólya–Schur multiplier sequence — whether it preserves hyperbolicity of *all* hyperbolic even polynomials, not just this family. The answer is no, by a one-line certificate.

Proposition 6.3 ((M_k) is not a multiplier sequence)**.** *The coefficientwise action $\sum_k a_k w^{2k} \mapsto \sum_k a_k M_k w^{2k}$ does not preserve hyperbolicity: the image of the hyperbolic polynomial $(w^2-1)^2$ is $M_2 w^4 - 2M_1 w^2 + 1$, all four of whose zeros are nonreal.*

Proof. The zeros satisfy $w^2 = \left(M_1 \pm \sqrt{M_1^2 - M_2}\right)/M_2$, and by (2), $M_0 = 1$, $M_1 = \mathrm{e}^{\gamma}/2$ and $M_2 = \frac{3}{4}\mathrm{e}^{2\gamma-1}$, so

$$M_1^2 - M_0 M_2 \;=\; \frac{\mathrm{e}^{2\gamma}}{4}\Big(1 - \frac{3}{\mathrm{e}}\Big) \;<\; 0 \qquad \text{because } \mathrm{e} < 3.$$

Hence w^2 is nonreal, and a nonreal number has no real square root. □

Remark 6.4 (moment sequences vs. multiplier sequences). The failure is structural, not an accident of (2). Applying the coefficientwise action of any sequence (γ_k) with $\gamma_k > 0$ to the hyperbolic polynomials $w^{2k-2}(w^2-1)^2$ for each $k \geq 1$ shows that an even multiplier sequence must satisfy the Turán inequalities $\gamma_k^2 \geq \gamma_{k-1}\gamma_{k+1}$, i.e. must be *log-concave*. But every moment sequence $\gamma_k = \mathbb{E}[V^k]$ of a nonnegative random variable is *log-convex* by Cauchy–Schwarz, strictly at $k = 1$ unless V is a.s. constant. So no nondegenerate moment sequence is a multiplier sequence — the two classes meet only in the geometric sequences $\gamma_k = c^k$, whose action is the rescaling $w \mapsto \sqrt{c}\,w$. In particular, no Pólya–Schur-type universal coefficientwise preserver theorem could prove Theorem 1.2: what the theorem certifies is a property of the *pair* (sequence, family), powered by the energy structure of C_d (Lemma 3.4) and the atom bound $p > \frac{1}{2}$ — and by Proposition 6.3 that pairing is essential. We regard this as making Theorem 1.2 more interesting, not less.

Acknowledgments. The mathematical route and its formal verification were developed jointly by two AI research agents, Claude ("Fable", Anthropic) and GPT (OpenAI), working as a coordinated pair under the author's direction, with mutual line-audits and the Lean kernel as referee. Three external referee passes by a third model (Gemini, Google) hardened earlier versions of the manuscript material.

References

[1] G. Szegő, *Orthogonal Polynomials*, 4th ed., Amer. Math. Soc. Colloq. Publ., vol. 23, American Mathematical Society, Providence, RI, 1975. ISBN 978-0-8218-1023-1.

[2] G. Pólya and J. Schur, *Über zwei Arten von Faktorenfolgen in der Theorie der algebraischen Gleichungen*, J. Reine Angew. Math. **144** (1914), 89–113. DOI: `https://doi.org/10.1515/crll.1914.144.89`.

[3] NIST Digital Library of Mathematical Functions, `https://dlmf.nist.gov/`, Release 1.2.6 of 2026-03-15; F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.

[4] E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis*, 4th ed., Cambridge University Press, 1927.

[5] J. F. C. Kingman, *Poisson Processes*, Oxford Studies in Probability, vol. 3, Oxford University Press, 1993. ISBN 978-0-19-853693-2.

[6] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci., vol. 12699, Springer, 2021, 625–635. DOI: `https://doi.org/10.1007/978-3-030-79876-5_37`.

[7] The mathlib Community, *The Lean mathematical library*, in: Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381. DOI: `https://doi.org/10.1145/3372885.3373824`.

Email address: `haufschildmichael@gmail.com`